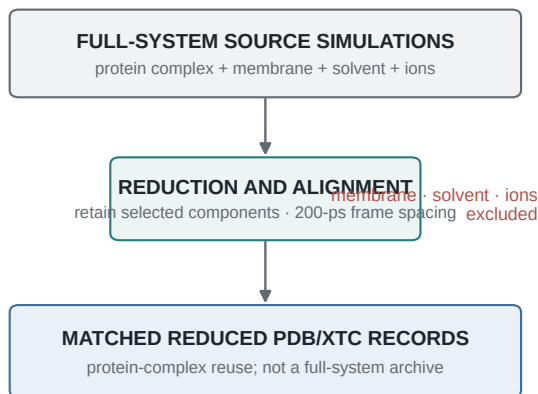


A

Molecular-record boundary



B

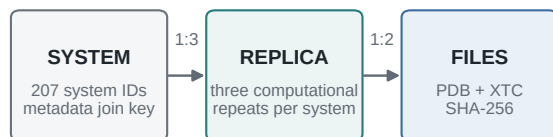
What each record can support

	protein	lipid	solvent ions	gateway recalc.
Reduced trajectory	Y	—	—	—
Gateway summary	S	—	—	Y
Full-system source	Y	Y	Y	Y

S = processed summary supplied; molecular recalculation still requires the source trajectories

C

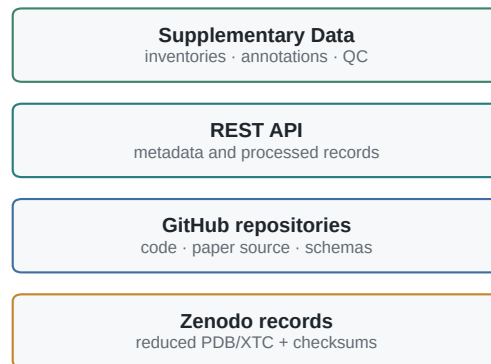
Replica-aware organization



File-level manifest joins sizes, checksums, atom counts, frame counts and component scope.

D

Access and reuse layers



Public-release claims remain conditional on unauthenticated DOI access.